

WHEN DOES CONSERVED-FLOW ROUTING HELP? A COUPLING CONDITION FOR MIXTURE-OF-EXPERTS, WITH A LEGALITY GUARANTEE

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ABSTRACT

Mixture-of-Experts (MoE) models route each input through a few experts chosen by a learned gate, one layer at a time and independently. We ask when a *coupled* router—one that commits to a single, globally consistent path of experts—is worth its cost. We study a conserved-flow router: a differentiable Tero–Nakagaki (Physarum) dynamic that relaxes to one min-cost path through a graph whose edges are exactly the legal expert transitions, so illegal routing is impossible by construction rather than merely penalized. Our central finding is a boundary condition. When a task’s output *marginalizes* over its intermediate steps—so each step can be scored independently—the coupling earns nothing: on MetaQA 2-hop knowledge-graph QA the flow router *ties* a plain per-step softmax over the same legal graph. As the steps become mutually constraining, the flow’s advantage over independent gating grows monotonically, a crossover we chart on a controlled constraint-density sweep. In a small MoE transformer on a coupled compositional task, the flow router cuts the illegal-transition rate from 0.34 to 0.01 at one dominant expert per layer; its accuracy reaches 0.99 on some seeds but is unstable across seeds and below a top-1 gate on average, which we report seed by seed. We also give the machinery that makes this practical: implicit differentiation through the flow fixed point and a matrix-free solver, verified against finite differences. The contribution is a mechanism, a precise condition for its use, and a hard legality guarantee—not a new accuracy record.

1 INTRODUCTION

Sparse Mixture-of-Experts (MoE) models (Shazeer et al., 2017; Fedus et al., 2022) scale capacity by activating only a few experts per input, chosen by a learned gate. The standard gate decides *independently* at every layer: it picks the best-scoring expert(s) for this layer without committing to a consistent route through the network, and nothing prevents it from chaining experts in an order the task forbids. A natural alternative is a *coupled* router that commits to one globally consistent path of experts. We study such a router built from a conserved flow: a differentiable Physarum dynamic (Tero et al., 2007; 2010) that relaxes to a single min-cost path through a graph whose edges are exactly the legal expert transitions, so illegal routing is impossible by construction.

But a mechanism is only useful if we know *when* to reach for it, and our main finding is a boundary condition. When a task’s output *marginalizes* over its intermediate steps—so each step can be scored independently—a conserved-flow router earns nothing over a plain per-step softmax on the same legal graph; MetaQA 2-hop knowledge-graph QA (Zhang et al., 2018) is such a task, and there the two *tie*. When the steps are mutually constraining, the conservation coupling (a shared flow budget forcing a globally consistent path) wins, and the margin grows with how coupled the steps are (Fig. 1). Contributions:

- **A characterized condition for coupled routing:** a conserved-flow router beats independent per-step gating *iff* the steps are coupled; we show the crossover on a controlled constraint-density sweep and predict the MetaQA 2-hop tie from it (§4).

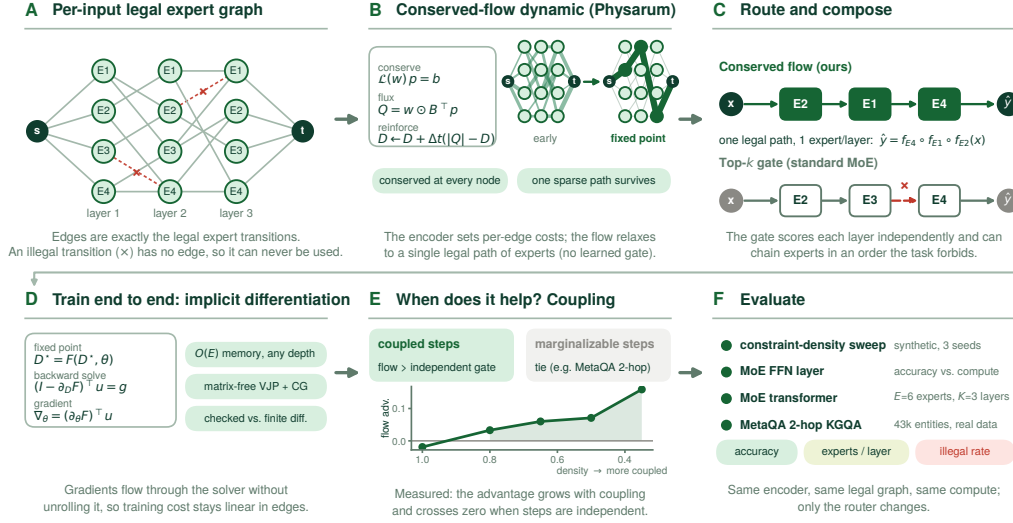


Figure 1: **Overview.** (A) Each input defines a graph whose edges are exactly the legal expert transitions; an illegal transition has no edge. (B) A conserved Physarum flow over this graph relaxes to one sparse min-cost path. (C) The model applies the experts on that path in sequence, whereas a standard top- k gate picks each layer independently and can chain experts illegally. (D) The flow is trained end to end by implicit differentiation through its fixed point. (E) The coupling condition: the flow beats an independent gate when steps constrain each other and ties when the output marginalizes over them (inset: measured advantage vs. legal-transition density, Fig. 3). (F) Testbeds and metrics.

- **A trainable MoE transformer, reported seed by seed:** on a coupled task the flow router cuts the illegal-transition rate by more than an order of magnitude relative to a top-1 gate on every seed; its accuracy is the best of any router on some seeds but unstable across seeds, and below the gate on average (Fig. 5).
- **Hard, input-conditional legality by topology:** the router’s flux is zero on every illegal transition, a guarantee rather than a penalty (§2).
- **An honest negative on MetaQA 2-hop:** on a marginalizable task the flow does not beat a matched no-flow baseline—evidence for the boundary, not against the mechanism.
- **Practical training:** implicit differentiation through the flow fixed point and a matrix-free solver, verified against finite differences (§3).

We do not claim to have invented the flow dynamic, which is the well-studied Tero–Nakagaki model; our contribution is its use as a trainable expert router, the legality guarantee, and the condition under which it helps.

2 METHOD

Graph and costs. A signed incidence $B \in \mathbb{R}^{N \times E}$ encodes the per-query legal graph. An encoder maps the input to per-edge logits z ; edge cost $L_e = \text{softplus}(-z_e) + \varepsilon$.

Conserved-flow dynamic. We use the Tero–Nakagaki model of *Physarum* transport networks (Nakagaki et al., 2000; Tero et al., 2007; 2010). With conductance D_e and weight $w_e = D_e/L_e$, flux $Q_e = w_e(B^\top p)_e$ obeys Kirchhoff conservation $\mathcal{L}(w)p = b$, where $\mathcal{L}(w) = B \text{diag}(w) B^\top$ is the weighted Laplacian and b injects $+1$ at the source and -1 at the sink (grounded). Reinforcement $D_e \leftarrow D_e + \Delta t(|Q_e| - D_e)$ iterates to the fixed point $D_e^* = |Q_e^*|$, at which a single sparse path survives.

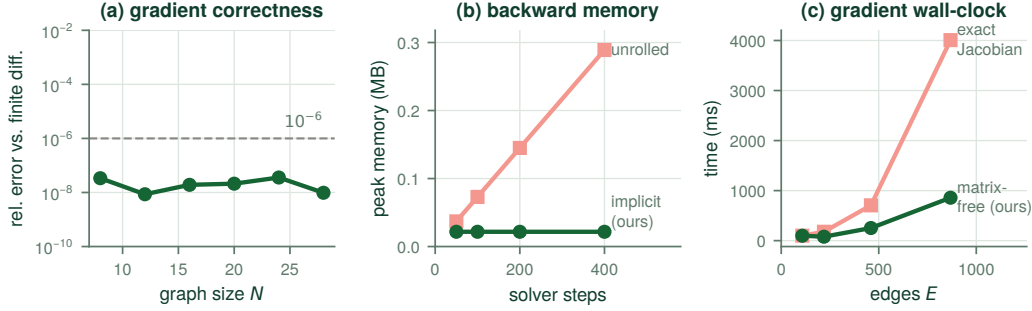


Figure 2: Differentiating and solving the flow at scale. (a) The implicit gradient matches a finite-difference gold standard to relative error $\sim 10^{-8}$ across graph sizes. (b) Backward memory is flat in solver depth for implicit differentiation (forward runs under `no_grad`) while unrolled autograd grows linearly. (c) The matrix-free (VJP-only) implicit gradient avoids the $O(E^2)$ Jacobian, so its wall-clock scales far better with edges.

Selection and composition. Node throughput $t_v = \frac{1}{2} \sum_e |B_{v,e}| |Q_e|$; per position $a_k = t^\gamma / \sum t^\gamma$. Walking the path, data nodes inject a value and expert nodes apply an MLP, giving function *composition* (not a weighted average).

Two guarantees. $e \notin E \Rightarrow Q_e = 0$ (legality by topology); and $\sum_{\text{in}} Q = \sum_{\text{out}} Q$ (conservation coupling).

3 SCALABILITY: DIFFERENTIATING AND SOLVING THE FLOW

The solve is one implicit layer $z^* = F(z^*, \theta)$ with $F(D) = |Q(D)|$, which we differentiate as in deep equilibrium models (Bai et al., 2019). By the implicit function theorem, for cotangent $g = \partial \mathcal{J} / \partial z^*$ the parameter gradient is $\partial \mathcal{J} / \partial \theta = (\partial F / \partial \theta)^\top u$ with $(I - (\partial F / \partial z)^\top) u = g$, solved matrix-free (VJP-only). This gives $O(E)$ backward memory independent of solver depth. We verify the implicit gradient against finite differences (cosine 1.0, rel-err $\sim 10^{-8}$) and against the exact Jacobian; a small conductance floor regularizes the otherwise-degenerate hard fixed point. A matrix-free conjugate-gradient forward solve (Laplacian-vector products from the edge list, $O(E)$ /iter) replaces the dense $O(N^3)$ solve, matching it to $\sim 10^{-3}$ and solving $N \approx 2400$ neighborhoods in ~ 0.4 s.

4 EXPERIMENTS

When does the flow help? (the central result). We sweep the density of a $K=3$ transition graph (3 seeds): low density = many illegal transitions = steps strongly coupled; density 1.0 = every transition legal = unconstrained/marginalizable. We compare the conserved flow against an *independent per-step softmax* over the identical legal graph (same encoder, same candidates; only the conservation solve differs). Figure 3 shows a clean monotonic crossover: the flow’s advantage is +0.159 MSE at the most coupled setting (density 0.35: flow 0.586 vs. independent 0.745) and falls to -0.019 when unconstrained (density 1.0: 0.509 vs. 0.490). The flow wins *iff* the steps are coupled.

MetaQA 2-hop: the predicted tie (real data). MetaQA (Zhang et al., 2018) KG: 43,234 entities, 9 relations, 134,741 triples; 2-hop questions, 8k train / 1.5k test, matched GRU encoder with per-hop query heads. A 2-hop answer marginalizes over the bridge entity, so the boundary predicts no benefit from coupling—and indeed the conserved flow ties a no-flow per-step softmax (Table 1). Both attain 0 hallucination by construction (the legal graph, not the flow, provides that).

As an MoE router (coupled regime). The same mechanism is a sparse expert router: the flow selects one legal *path of experts* and composes them sequentially. On a constrained sequential expert-composition task (density ≈ 0.6 , 4 seeds), the conserved flow beats both standard top- k

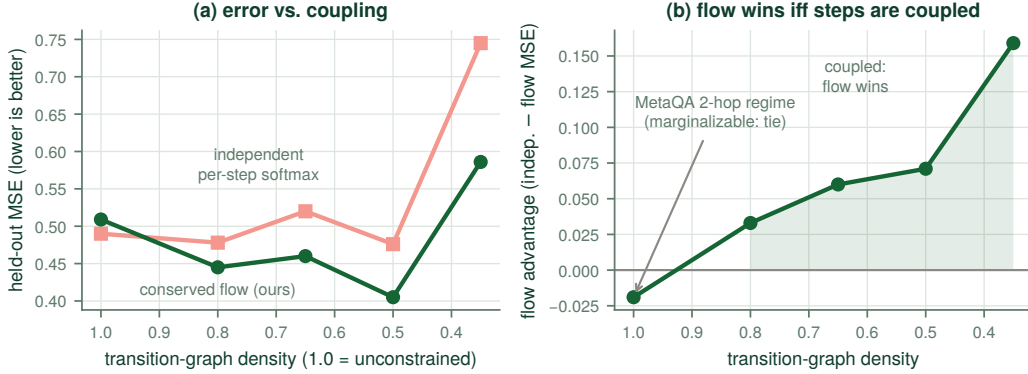


Figure 3: Conserved-flow routing beats independent per-step scoring exactly when reasoning steps are coupled. (a) held-out MSE vs. constraint density (left = more coupled); (b) the flow’s advantage crosses zero as the task becomes marginalizable. MetaQA 2-hop sits at the unconstrained end, predicting the tie we observe below.

Method	hits@1 (raw)	hits@1 (degree-norm)	hallucination
Conserved flow (ours)	0.864	0.948	0 (by construction)
No-flow ablation (same encoder/graph)	0.885	0.944	0 (by construction)

Table 1: MetaQA 2-hop. The conserved flow does not improve accuracy over a matched no-flow baseline on this marginalizable task—consistent with Fig. 3. The legality guarantee comes from the topology and holds for both. By contrast, embedding-based KGQA methods that score answers over *all* entities (e.g. EmbedKGQA (Saxena et al., 2020), built on ComplEx (Trouillon et al., 2016)) carry no such guarantee and can answer outside the legal set.

gating and Routing-Free MoE on accuracy and on legality (Table 2)—the coupled regime where Fig. 3 predicts a win.

Real MoE FFN layer: the accuracy/compute frontier. We stack K MoE FFN layers over $E=8$ experts and compare the conserved-flow router against a hard top- k gate (standard MoE with load balancing) and a dense layer, in the coupled regime (3 seeds; Fig. 4). The flow router is *Pareto-dominant*: at the minimum compute of one active expert per layer it attains the best held-out MSE (1.170 ± 0.033), beating the top-1 gate at the same compute (1.294 ± 0.033) and even the dense layer (1.278 ± 0.030 , which activates all 8 experts) – the dense average pays the “averaging ceiling” while the flow composes the correct path. On legality, the flow’s flux is supported only on legal expert transitions by construction; decoding its connected max-flux path is legal by construction, and even a per-layer argmax decode violates legality only 0.150 ± 0.010 of the time versus 0.899 for the top-1 gate. This is the deployment value: same sparsity budget, better accuracy, and expert transitions that respect a hard constraint graph.

A real MoE transformer. We move from an isolated FFN block to a small transformer (token embedding, multi-head attention, and an $E=6$ -expert MoE feed-forward block at each of $K=3$ layers) trained end-to-end on a coupled compositional sequence task, where the correct expert at a layer depends on the expert chosen at the previous layer (30k training sequences, 20 epochs, one run per seed). Because the flow’s behavior varies across seeds, we show every seed (Fig. 5) as well as the summary (Table 3).

Legality holds on every seed. The flow’s illegal-transition rate is 0.007 on average versus 0.336 for the top-1 gate, and never exceeds 0.034 on any seed. The residual nonzero rate is a decoding artifact: we read the route with a per-layer argmax of node throughput, which can pair experts from two different legal paths when the flow has not fully collapsed onto one; the flow’s flux itself is zero on every illegal edge, and decoding the connected max-flux path is legal by construction.

Router	held-out MSE $\downarrow$	illegal-transition rate $\downarrow$
Top- k gate (standard MoE)	0.506 ± 0.031	0.295 ± 0.021
Routing-Free MoE	0.583 ± 0.024	0.262 ± 0.019
Conserved flow (ours)	0.392 ± 0.026	0.103 ± 0.007
oracle (upper bound)	0.001	0.000

Table 2: Constrained sequential expert routing, 4 seeds (MSE / illegal-transition rate at noise 1.0). The conserved-flow router beats top- k gating and Routing-Free MoE on both, in the coupled regime. Illegal-transition rate is 0.000 at lower noise for the flow.

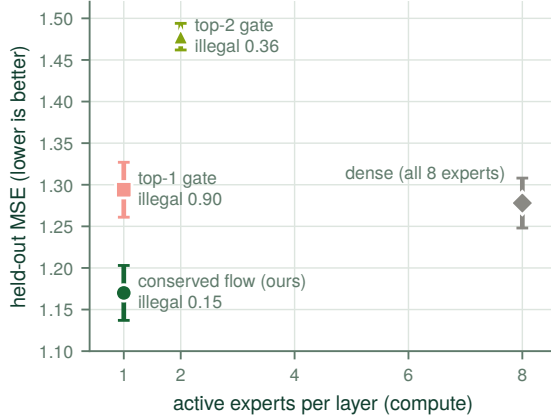


Figure 4: Conserved-flow routing is Pareto-dominant on the MoE accuracy/compute frontier (coupled regime, 3 seeds): lowest error at one active expert per layer, while the top- k gate needs more experts or routes illegally (illegal-transition rate annotated).

Accuracy is high-variance. When training succeeds, the flow is the most accurate router we tested (up to 0.992, versus a top-1 mean of 0.847), and it beats the top-1 gate on 5 of 10 seeds. On the remaining seeds it trains to a worse solution, in one case failing entirely, so its mean (0.756; median 0.820) is below the gate’s. We attribute this to instability of the inner solve early in training, when edge costs are near-uniform and the reinforcement dynamic can lock onto a wrong path; stabilizing it (for example by annealing the conductance floor or the sharpening exponent γ) is the most important open problem for this router. The gates are not stable at this scale either (top-1 falls to 0.45 on one seed and top-2 ranges from 0.11 to 0.99), and the dense layer, which averages all six experts, fails on this compositional task.

Legality wedge. A PathMoE-style *shared-router* baseline (router params tied across steps) attains the *worst* legality under noise (illegal-hop ≈ 0.5) versus the flow’s ≈ 0 : soft/param-sharing constraints can make legality worse, not merely fail to guarantee it—the safety argument for topology-enforced expert routing in deployed MoE.

5 RELATED WORK

Sparse MoE routing. Standard MoE (Shazeer et al., 2017; Fedus et al., 2022) uses a learned top- k gate per layer, chosen *independently*; load-balancing losses keep experts utilized. PathMoE (Gu et al., 2026) constrains the path space by *tying* router parameters across positions (a soft, statistical constraint); Chain-of-Experts (Wang et al., 2025) chains experts sequentially within a layer with a learned router at each step; Routing-Free MoE (Liu et al., 2026) drops the gate for threshold self-activation. All route by learned scores with no hard guarantee on which expert may follow which. Our router is a conserved flow whose *topology* makes illegal expert transitions impossible, whose objective is a min-cost path (not a sampler), and which is differentiable and jointly trained; our

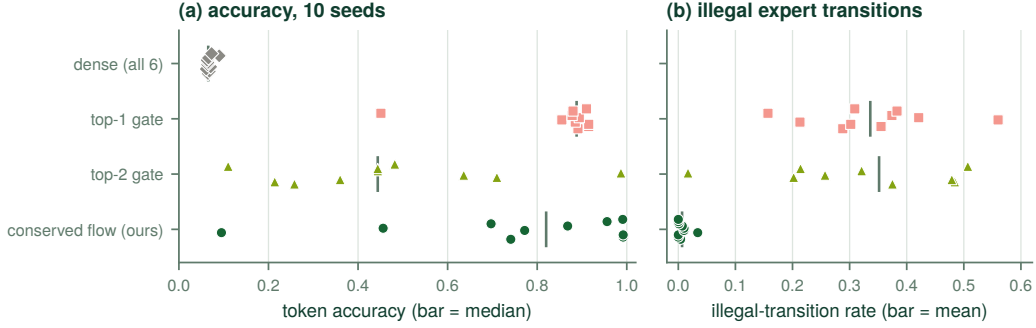


Figure 5: MoE transformer on a coupled compositional task, one point per seed. (a) Token accuracy: the flow router reaches the highest accuracy of any router on some seeds but is unstable across seeds. (b) Illegal expert transitions: the flow stays near zero on every seed while both top- k gates route illegally on a large fraction of inputs.

Router	token acc $\uparrow$	median acc	experts/layer	illegal-transition $\downarrow$
dense (all 6)	0.069 ± 0.008	0.066	6	—
top-1 gate	0.847 ± 0.133	0.888	1	0.336 ± 0.106
top-2 gate	0.464 ± 0.246	0.444	2	0.352 ± 0.178
conserved flow (ours)	0.756 ± 0.274	0.820	$1^\dagger$	0.007 ± 0.010

Table 3: MoE transformer, mean $\pm$ standard deviation over 10 seeds (per-seed values in Fig. 5).

† The flow layer weights expert outputs by node throughput, which concentrates on one expert per layer at convergence; the top- k gates apply exactly k .

contribution over these is (i) the hard, input-conditional legality guarantee and (ii) the *condition* (coupling) under which such routing beats an independent gate.

Flow / graph reasoning. GFlowNets (Bengio et al., 2021) (flow *sampling*, not min-cost); PathRAG (Chen et al., 2026) (non-parametric flow pruning); D-RAG (Gao et al., 2025) (differentiable subgraph *set*); MoG (Yuan et al., 2026) (a topology-aware router over expert graphs); Physarum differentiable-LP layers (Meng et al., 2021) (LPs, not expert routing). We concede the scooped *outcomes* (PathMoE’s path-constraint benefits, CoE’s sequential composition, GFlowNet’s flow-over-DAG) and claim the mechanism, the guarantee, and the coupling condition.

6 LIMITATIONS

The flow’s benefit is conditional: on marginalizable tasks it does not beat independent scoring, so the mechanism should be reached for only when reasoning steps are coupled (the contribution is precisely this condition). The legality source is the KG (clean on KGQA, open where no native graph exists); γ and the conductance floor are tuned by hand. In the MoE transformer the flow’s accuracy is unstable across seeds (Fig. 5), and its layer weights experts by throughput rather than applying exactly one, so a strict hard-path evaluation at equal compute is still needed. The coupled-regime win is shown on controlled synthetic constraints and on MetaQA it is the tie we predict—a real-data *coupled* instance (e.g. 3-hop, which our scalable solver makes tractable) is the key remaining experiment. Results are single-domain (movies); WebQSP/CWQ are needed for external validity.

7 CONCLUSION

A conserved-flow router commits a Mixture-of-Experts model to one globally consistent, min-cost path of experts, and its topology makes illegal expert transitions impossible by construction. We asked not whether such routing works but *when it is worth it*, and found a clean boundary: when a task’s output marginalizes over its intermediate steps, the coupling buys nothing and an independent

per-step gate suffices (MetaQA 2-hop, a tie); when the steps are mutually constraining, the flow’s advantage over independent gating grows with the coupling. In a trainable MoE transformer the flow keeps illegal routing near zero on every seed and reaches the best accuracy of any router when it trains well, but its training is not yet reliable. Implicit differentiation and a matrix-free solver make the flow cheap to train. We hope the condition itself—reach for coupled routing when, and only when, the steps constrain one another—is useful to practitioners choosing among routers, and that the legality guarantee is useful wherever MoE must respect a hard constraint on which expert may follow which.

AI USE STATEMENT

The research direction and the central idea (using a conserved Physarum flow as an expert router, and asking when coupled routing helps) originated with the author. We used generative AI tools (an LLM-based coding assistant) extensively: to implement and debug the experiment code, to run experiments, to produce the figures, to search for and verify related work, and to draft and edit the text of this paper. We have reviewed all AI-assisted work: every number reported here comes from code in our repository, the implicit-differentiation code is checked against finite differences, and every reference was checked against the published paper. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

All experiments use small models and public data (MetaQA; Zhang et al., 2018) and run on a single GPU or CPU. The method is specified in §2 and §3; for every table we report the number of seeds and mean $\pm$ standard deviation, and the implicit gradient is verified against finite differences (Fig. 2). Anonymized code for all experiments and figures, with the raw per-seed logs behind every table, is included in the supplementary material and will be released publicly upon publication.

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